BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5483

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The atomic number of sodium is:

(A) 3
(B) 8
(C) 9
(D) 11

سوڈیم کا ایٹمک نمبر ہے:

Q2. A weak electrolyte is:

(A) NaOH
(B) CH_3COOH
(C) Na_2CO_3
(D) $C_6H_{12}O_6$

کمزور الیکٹرولائٹ ہے:

Q3. The formula of rust is:

(A) Fe_2O_3
(B) $Fe_2O_3 \cdot nH_2O$
(C) $Fe(OH)_3 \cdot nH_2O$
(D) $Fe(OH)_3$

زنگ کا فارمولا ہے:

Q4. A universal solvent on the Earth is:

(A) Ether
(B) Alcohol
(C) Ammonia
(D) Water

زمین پر یونیورسل سالونٹ ہے:

(A) ایتھر
(B) الکوحل
(C) امونیا
(D) پانی

Q5. The constant factor in Boyl's law is:

(A) Temperature
(B) Volume
(C) Pressure
(D) Mole

بوائل کے قانون میں مستقل مقدار ہے:

(A) درجہ حرارت
(B) حجم
(C) پریشر
(D) مول

Q6. The blood pressure of healthy young man is:

(A) 100/80
(B) 120/80
(C) 150/80
(D) 160/80

صحت مند نوجوان انسان کا بلڈ پریشر ہوتا ہے:

Q7. How many covalent bonds in N_2 molecule?

(A) 3
(B) 4
(C) 5
(D) 6

N_2 مالیکیول میں کتنے کوویلنٹ بانڈز ہوتے ہیں؟

Q8. The boiling point of sodium chloride is:

(A) 1413°C
(B) 1313°C
(C) 1213°C
(D) 1113°C

سوڈیم کلورائیڈ کا بوائلنگ پوائنٹ ہے:

Q9. Electronegativity of fluorine is:

(A) 1
(B) 2
(C) 3
(D) 4

فلورین کی الیکٹرونیگیٹیویٹی ہے:

Q10. The value of 'n' for M shell is:

(A) 1 (B) 2

(C) 3 (D) 4

M شیل کے لیے 'n' کی قیمت ہے:

Q11. The molar mass of H_2SO_4 is:

(A) 98 gm (B) 98 amu

(C) 9.8 gm (D) 9.8 amu

H_2SO_4 کا مولر ماس ہے:

Q12. The third abundant gas found in the Earth's atmosphere:

(A) Nitrogen (B) Argon

(C) Oxygen (D) Hydrogen

زمین کی فضا میں کثرت کے لحاظ سے تیسرے نمبر پر کونسی گیس پائی جاتی ہے؟

(A) نائٹروجن **(B)** آرگان

(C) آکسیجن **(D)** ہائیڈروجن

Chemistry (Subjective) Group - I

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5483

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define physical properties and give example. (1+1)

(ii) Write the names of any two elements which are found in solid and liquid states. (1+1)

(iii) What is the difference between empirical and molecular formulae? (1+1)

(iv) How Neutrons were discovered?

(v) How isotopes are used in power generations?

(vi) Define Periodic Law.

(vii) What is the difference between lanthanides and actinides? (1+1)

(viii) Define electronegativity and give example. (1+1)

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) What are ionic compounds? Give an example.

(ii) What is a single covalent bond? Give an example.

(iii) What are polar compounds? Give an example.

(iv) Define evaporation. Write its one advantage.

(v) What are amorphous solids?

(vi) Define standard atmospheric pressure.

(vii) Write balanced chemical equation for the reaction of O_2 with alkali metals.

(viii) Write any two uses of platinum.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Why is water called universal solvent?

(ii) What is the difference between dilute and concentrated solution?

(iii) How 1M solution of NaOH is prepared?

(iv) What is meant by % m/v? Explain.

(v) Calculate oxidation number of nitrogen in HNO_3.

(vi) Define oxidation in terms of electrons.

(vii) What is galvanizing? Write its advantage. (1+1)

(viii) In the reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ which atom is oxidized and reduced?

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5 (5×1). (Marks 5×1) Write any five major characteristics of cathode rays.

Q5 (4×1). (Marks 4×1) Explain different types of molecules with examples.

Q5 (3+2). (Marks 3+2) Define ionic bond. Explain it with an example.

Q5 (2+2). (Marks 2+2) State Boyle's Law. Describe its mathematical explanation.

Q6 (5×1). (Marks 5×1) Discuss manufacturing of NaOH from brine in Nelson's cell.

Q6 (4×1). (Marks 4×1) Write four important chemical properties of metals.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5488

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Which compound has double covalent bond?

(A) C_2H_2 (B) C_2H_4

(C) CH_4 (D) N_2

ذیل کو ویلنٹ بانڈ والا کونزا ہے؟

Q2. A bond formed between two non-metals is expected to be:

(A) Covalent (B) Coordinate covalent

(C) Ionic (D) Metallic

دو نان میٹل کے درمیان بننے والا بانڈ متوقع طور پر ہے:

(A) کو ویلنٹ **(B)** کوآرڈینیٹ کو ویلنٹ

(C) آئیونک **(D)** میٹلک

Q3. Along the period, which one decreases?

(A) Electron affinity (B) Electro-negativity

(C) Ionization energy (D) Atomic radius

ایک پیریڈ میں بائیں سے دائیں کونسی کم ہوتی جاتی ہے؟

(A) الیکٹران افینیٹی **(B)** الیکٹرو نیگیٹیویٹی

(C) آئیونائزیشن انرجی **(D)** ایٹمک ریڈیس

Q4. Which isotope is used for treatment of skin cancer?

(A) Co-60 (B) Sr-90

(C) C-14 (D) I-131

جلد کے کینسر کے علاج میں کونزا آئسوٹوپ استعمال ہوتا ہے؟

Q5. Which compound can be separated by physical means?

(A) Element
(B) Compound
(C) Mixture
(D) Radical

کس کے اجزاء کو طبعی طریقوں سے الگ الگ کیا جا سکتا ہے؟

(A) ایلیمنٹ
(B) کمپاؤنڈ
(C) مکسچر
(D) ریڈیکل

Q6. The removal of an electron from an atom gives:

(A) Anion
(B) Cation
(C) Molecule
(D) Molecular ion

ایٹم سے ایک الیکٹران کا اخراج بناتا ہے:

(A) اینائن
(B) کیٹائن
(C) مالیکیول
(D) مالیکیولر آئن

Q7. Which non-metal is lustrous?

(A) Sulphur
(B) Phosphorus
(C) Carbon
(D) Iodine

کونسی نان میٹل چمکدار ہے؟

(A) سلفر
(B) فاسفورس
(C) کاربن
(D) آیوڈین

Q8. The oxidation number of oxygen in OF_2 is:

(A) -1
(B) +1
(C) -2
(D) +2

OF_2 میں آکسیجن کا آکسیڈیشن نمبر ہے:

Q9. Which is a weak electrolyte?

(A) Sulphuric acid
(B) Acetic acid
(C) Sodium chloride
(D) Sodium hydroxide

کونسا کمزور الیکٹرولائٹ ہے؟

(A) سلفیورک ایسڈ
(B) ایسیٹک ایسڈ
(C) سوڈیم کلورائیڈ
(D) سوڈیم ہائیڈرو آکسائیڈ

Q10. If a solute in cm^3 is dissolved in $100cm^3$ of the solution, the percentage is:

(A) % m/v
(B) % m/m
(C) % v/m
(D) % v/v

اگر سولیوٹ کے $3cm^3$ میں $100cm^3$ محلول میں حل کیا جائے تو شرح ہوتی ہے:

Q11. Identify the transition temperature of tin:

(A) 96°C
(B) 91°C
(C) 13.2°C
(D) 230°C

ٹن کے ٹرانزیشن ٹمپریچر کی شناخت کیجیے:

Q12. Vapour pressure of water in mmHg at 60°C is:

(A) 17.5
(B) 55.3
(C) 149.4
(D) 355.1

60°C پر پانی کا ویپر پریشر mmHg میں ہوتا ہے:

Chemistry (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5488

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define organic chemistry.

(ii) Write two difference between atom and ion. (1+1)

(iii) What is molecular mass? Give its example. (1+1)

(iv) Write any two defects of Rutherford's Atomic Model. (1+1)

(v) Define isotope and give an example. (1+1)

(vi) Define ionization energy and write its trend in group. (1+1)

(vii) Why the elements of group I and group II are called s-block elements?

(viii) Define shielding effect.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Define single covalent bond with one example. (1+1)

(ii) What are intermolecular forces?

(iii) What is hydrogen bonding?

(iv) Why density of gases increases on cooling?

(v) Which principle is used in working of pressure cooker?

(vi) What is transition temperature? Explain it with one example. (1+1)

(vii) Write two uses of gold.

(viii) Describe the trend of electropositivity in periodic table. (1+1)

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) What is the difference between solute and solvent? (1+1)

(ii) What is %age m/m? Write its formula. (1½+½)

(iii) Define solubility and give an example. (1½+½)

(iv) Define false solution and give an example. (1½+½)

(v) Define oxidation and reduction in the term of electrons. (1+1)

(vi) What are weak electrolytes? Give example. (1½+½)

(vii) Why galvanizing is done?

(viii) What do you mean by tin electroplating?

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5 (5×1). (Marks 5×1) Describe the five postulates in Rutherford's Model.

Q5 (3+1). (Marks 3+1) What is a molecule? Discuss its three types.

Q5 (3+2). (Marks 3+2) Define hydrogen bonding. Explain that how these forces affect the physical properties of compounds?

Q5 (2+2). (Marks 2+2) State and explain Charles's Law and derive its equation.

Q6 (5×1). (Marks 5×1) Describe any five rules for assigning oxidation number.

Q6 (4×1). (Marks 4×1) How does H_2 react with the following? F_2, Cl_2, Br_2, I_2